

Practical No.3. Implementation of DML,DCL and TCL Commands.

1) DML (Data Manipulation Language)

A) Insert

```
insert into Student(Name,Roll_no,Address,Branch)      values( "Raj
pandey",1,"A/P Bhor Tal-Bhor Dist-Pune", "Computer"),
("Krish Kapoor",2,"A/P Wadi Tal-Bhor Dist-Pune", "Computer"),
("Ram Kukreja",3,"A/P Velu Tal-Bhor Dist-Pune", "AIML");
```

```
mysql> Select* from Student;
+-----+-----+-----+-----+
| Name          | Roll_no | Address                                     | Branch |
+-----+-----+-----+-----+
| Raj pandey    | 1       | A/P Bhor Tal.Bhor Dist.Pune | AIDS   |
| Krish Kapoor | 2       | A/P Wadi Tal.Bhor Dist.Pune | Computer |
| Ram Kukreja   | 3       | A/P Velu Tal.Bhor Dist.Pune | AIML    |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

B) Update

```
update Student      set
Name="Radha Kapoor"
where Branch="AIDS";
```

```
mysql> SELECT * FROM Student;
+-----+-----+-----+-----+
| Name          | Roll_no | Address                                     | Branch |
+-----+-----+-----+-----+
| Radha Kapoor  | 1       | A/P Bhor Tal.Bhor Dist.Pune | AIDS   |
| Krish Kapoor  | 2       | A/P Wadi Tal.Bhor Dist.Pune | Computer |
| Ram Kukreja   | 3       | A/P Velu Tal.Bhor Dist.Pune | AIML    |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

C) Delete

```
delete from Student
where Roll_No=3;
```

```
mysql> select* FROM Student;
+-----+-----+-----+-----+
| Name          | Roll_no | Address                                     | Branch |
+-----+-----+-----+-----+
| Radha Kapoor  | 1       | A/P Bhor Tal.Bhor Dist.Pune | AIDS   |
| Krish Kapoor  | 2       | A/P Wadi Tal.Bhor Dist.Pune | Computer |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

2) (DCL)Data Control Language

A)Create user

mysql> create user "student_user"@"localhost"
-> identified by "pass@123";

```
mysql> select user, host from mysql.user;
+-----+-----+
| user          | host          |
+-----+-----+
| debian-sys-maint | localhost    |
| mysql.session  | localhost    |
| mysql.sys      | localhost    |
| root           | localhost    |
| student_user   | localhost    |
+-----+-----+
5 rows in set (0.00 sec)
```

B)Grant mysql> grant select, insert, update, delete -> on
mysql.Student
-> to "student_user"@"localhost";

mysql> update Student
-> set fees_paid=23000
-> where Roll_no=1;

mysql> update Student
-> set fees_paid=53000
-> where Roll_no=2;

```
mysql> select * from Student;
+-----+-----+-----+-----+-----+
| Name          | Roll_no | Address                                     | Branch   | fees_paid |
+-----+-----+-----+-----+-----+
| Radha Kapoor  | 1       | A/P Bhore Tal.Bhore Dist.Pune | AIDS     | 23000     |
| Krish Kapoor  | 2       | A/P Wadi Tal.Bhore Dist.Pune | Computer | 53000     |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

C)Revoke

mysql> revoke select
-> on mysql.Student
-> from "student_user"@"localhost";

3) TCL(Transaction Control Language)

1) Commit

- Check autocommit is on.

```
mysql> show variables like "autocommit";
+-----+-----+
| Variable_name | Value |
+-----+-----+
| autocommit    | ON    |
+-----+-----+
1 row in set (0.00 sec)
```

- Start Transaction. mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

```
mysql> update Student
-> set fees_paid=fees_paid+7000
-> where Roll_no=1;
Query OK, 1 row affected (0.00 sec)
```

```
mysql> select * from Student;
+-----+-----+-----+-----+-----+
| Name      | Roll_no | Address                                     | Branch | fees_paid |
+-----+-----+-----+-----+-----+
| Radha Kapoor |      1 | A/P Bhore Tal.Bhore Dist.Pune | AIDS   |    30000 |
| Krish Kapoor |      2 | A/P Wadi Tal.Bhore Dist.Pune | Computer |    53000 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

2) Rollback mysql->rollback;